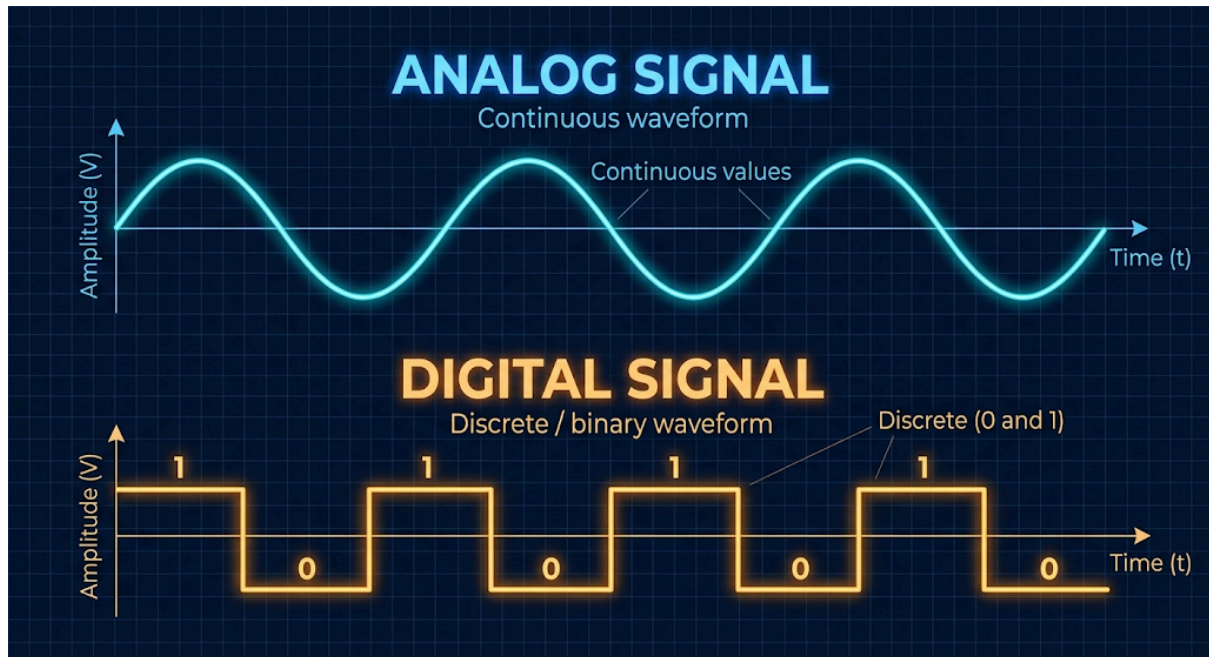


Analog Signal

An **analog signal** is a continuous wave that varies constantly over time to represent information, like human voice or natural sound.

Digital Signal

A **digital signal** is a discrete wave that represents data using binary code (specifically zeros and ones, 0 and 1) like computer data or digital audio files.



	ANALOG SIGNAL	DIGITAL SIGNAL
Signal Form	Continuous wave	Discrete pulses (0,1)
Noise	Accumulates and hard to repair	Regenerated and noise removed
Distance	Short repeater spacing	Much longer distance
Security	Low Security	High Security
Repeater	Amplify (signal + noise)	Regenerate original signal
Management	Poor	Good
BER (bit error rate)	10^{-5}	$10^{-7} - 10^{-11}$